Circles and Parabolas

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circles**

Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

$(x-3)^2 + (y+2)^2 = 25$.

2 **[Foundational]** • **Target: Circles**

Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

a Write the equation in standard form by completing the square. $(x-3)^2 + (y+2)^2 = 25$.

b State the center and radius. Center $(3, -2)$, radius 5.

(a) $(x-3)^2 + (y+2)^2 = 25$.; (b) Center $(3, -2)$, radius 5.

3 **[Foundational]** • **Target: Circles**

Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x-3)^2 + (y+2)^2 = 25$.

$(6-3)^2 + (2+2)^2 = 9 + 16 = 25$, so the point lies on the circle.

4 **[Foundational]** • **Target: Parabolas with vertex at the origin**

For the parabola $y^2 = 8x$, find:

a the focus $(2, 0)$

b the directrix $x = -2$

c the direction of opening opens right

(a) $(2, 0)$; (b) $x = -2$; (c) opens right

5 **[Foundational]** • **Target: Parabolas with vertex at the origin**

A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.

a Find its equation in the form $x^2 = 4py$. $16 = 4p(2)$ gives $p = 2$, so $x^2 = 8y$.

b State the coordinates of the focus. $(0, 2)$.

(a) $16 = 4p(2)$ gives $p = 2$, so $x^2 = 8y$.; (b) $(0, 2)$.

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6 [Foundational] • Target: Parabolas with vertex at the origin

A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).

The edge point $(1, 0.5)$ lies on $x^2 = 4py$: $1 = 4p(0.5) = 2p$, so $p = 0.5$ m. The receiver sits 0.5 m above the vertex.

7 [Foundational] • Target: Parabolas with vertex not at the origin

For the parabola $(y - 2)^2 = 12(x + 1)$, find:

a the vertex $(-1, 2)$

b the value of p $4p = 12$, so $p = 3$

c the focus $(2, 2)$

d the directrix $x = -4$

(a) $(-1, 2)$; (b) $4p = 12$, so $p = 3$; (c) $(2, 2)$; (d) $x = -4$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parabolas with vertex not at the origin

Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.

Opening downward: $(x - 2)^2 = -4(y + 3)$.

9 [Intermediate] • Target: Parabolas with vertex not at the origin

Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

$y = (x - 3)^2 - 4$. Vertex: $(3, -4)$.

10 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.

Substitute $x^2 = y + 5$ into the circle: $(y + 5) + y^2 = 25$, so $y^2 + y - 20 = 0$, giving $(y + 5)(y - 4) = 0$: $y = -5$ or $y = 4$. For $y = -5$: $x^2 = 0$, point $(0, -5)$. For $y = 4$: $x^2 = 9$, points $(\pm 3, 4)$. Three intersection points: $(0, -5)$, $(3, 4)$, $(-3, 4)$.

11 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

Radius $= 6400 + 500 = 6900$ km.

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12 [Intermediate] • Target: Graphing circles and parabolas

The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.

From the graph, x-intercepts near $x = -1, 3$ and vertex near $(1, -4)$. Algebraically: $y = (x-1)^2 - 4$, vertex $(1, -4)$; setting $y = 0$: $(x-1)^2 = 4 \Rightarrow x = -1, 3$ ✓.

13 [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

Radius slope from origin to $(3, 4)$: $\frac{4}{3}$. Tangent slope (perpendicular): $-\frac{3}{4}$. Tangent line: $y - 4 = -\frac{3}{4}(x - 3)$, i.e. $y = -\frac{3}{4}x + \frac{25}{4}$.

14 [Intermediate] • Target: The reflective property of parabolas

Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Any ray traveling parallel to a parabola's axis of symmetry reflects off the interior surface and passes through the focus (or, conversely, rays originating at the focus reflect out parallel to the axis) — this is why satellite dishes concentrate incoming signals at the focus, and headlights placed at the focus project a parallel beam.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

Substitute each of the three points into the general circle equation $x^2 + y^2 + Dx + Ey + F = 0$, producing a system of three linear equations in D, E, F. Solving this system gives the circle's equation, which can then be converted to standard form by completing the square.

16 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 9.02 Geometric Sequences and Series

State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

A parabola is the set of points equidistant from the focus and directrix. At $(4, 4)$: distance to focus $(1, 0)$ is $\sqrt{(4-1)^2 + 4^2} = \sqrt{9 + 16} = 5$. Distance to directrix $x = -1$ is $4 - (-1) = 5$. Equal, confirming the point lies on the parabola.

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17 [Challenging] • Target: Systems involving circles • Cross-Lesson: 9.03 The Binomial Theorem

Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x-6)^2 + y^2 = 25$.

Subtracting the equations: $x^2 - (x-6)^2 = 0 \Rightarrow x^2 - (x^2 - 12x + 36) = 0 \Rightarrow 12x - 36 = 0 \Rightarrow x = 3$. Substituting into the first: $9 + y^2 = 25 \Rightarrow y = \pm 4$. Intersection points: $(3, 4)$ and $(3, -4)$.

18 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 1.01 Functions and Function Notation

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2-1, 4-2, 1-3 \rangle = \langle 1, 2, -2 \rangle$. $\vec{AC} = \langle -1-1, 0-2, 5-3 \rangle = \langle -2, -2, 2 \rangle$. Step 2: Compute normal vector via cross product: $\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 2, -2; -2, -2, 2| = \mathbf{i}(4-4) - \mathbf{j}(2-4) + \mathbf{k}(-2-(-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2 \rangle$. Simplifying by scalar.

19 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 1.02 Domain and Range

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of $b = 3$ back into E.

20 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 1.03 Average Rate of Change

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Step 1: Identify ellipse parameters: The vertices and foci lie on the x-axis, so the major axis is horizontal and centered at $(0, 0)$. Here semi-major axis $a = 5$ and focal distance $c = 3$. Step 2: Compute semi-minor axis b: For an ellipse: $b^2 = a^2 - c^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow b = 4$. Step 3: Formulate equation and invariants: $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Major axis length $2a = 10$, minor axis length $2b = 8$, and eccentricity $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Standard Equation and Features of Circles

Find the center and radius of the circle $(x-3)^2+(y+5)^2=49$.

a. **Center:** $(3,\ -5)$**, Radius:** 7 ✓

b. Center: $(-3, 5)$, Radius: 7

c. Center: $(3,\ -5)$, Radius: 49

d. Center: $(-3, 5)$, Radius: 49

Standard form is $(x-h)^2+(y-k)^2=r^2$. Here $h=3, k=-5, r=\sqrt{49}=7$.

22 [Foundational] • Target: Standard Equation and Features of Circles

Convert the general equation $x^2+y^2-6x+8y-11=0$ to standard form by completing the square.

a. $(x-3)^2+(y+4)^2=36$ ✓

b. $(x+3)^2+(y-4)^2=36$

c. $(x-3)^2+(y+4)^2=25$

d. $(x-6)^2+(y+8)^2=11$

$(x^2-6x+9)+(y^2+8y+16)=11+9+16=36 \Rightarrow (x-3)^2+(y+4)^2=36$.

23 [Foundational] • Target: Standard Equation and Features of Circles

Find the equation of the circle passing through the origin with center $(-2, 3)$.

a. $(x+2)^2+(y-3)^2=13$ ✓

b. $(x-2)^2+(y+3)^2=13$

c. $(x+2)^2+(y-3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

Radius $r=\sqrt{(-2-0)^2+(3-0)^2}=\sqrt{4+9}=\sqrt{13}$. Equation: $(x+2)^2+(y-3)^2=13$.

24 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the focus and directrix of the parabola $x^2=12y$.

a. **Focus:** $(0, 3)$**, Directrix:** $y=-3$ ✓

b. Focus: $(3, 0)$, Directrix: $x=-3$

c. Focus: $(0, 12)$, Directrix: $y=-12$

d. Focus: $(0, 6)$, Directrix: $y=-6$

$x^2=4py \Rightarrow 4p=12 \Rightarrow p=3$. Vertical axis, opens up: Focus is $(0, p)=(0, 3)$, directrix is $y=-p=-3$.

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25 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the standard equation of the parabola with vertex $(0, 0)$ and focus $(0, -2)$.

a. $x^2 = -8y$ ✓

b. $y^2 = -8x$

c. $x^2 = 8y$

d. $y^2 = -4x$

Opens downward: $p = -2$. Standard equation: $x^2 = 4py = 4(-2)y = -8y$.

26 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the focal diameter (latus rectum length) of $(x - 1)^2 = -16(y + 2)$.

a. 16 ✓

b. 4

c. 8

d. 32

The focal diameter is $|4p| = |-16| = 16$.

27 [Foundational] • Target: Parabolas with Horizontal Axis of Symmetry

Find the vertex, focus, and directrix of $(y - 2)^2 = 8(x + 1)$.

a. **Vertex:** $(-1, 2)$**, Focus:** $(1, 2)$**, Directrix:** $x = -3$ ✓

b. Vertex: $(1, -2)$, Focus: $(3, -2)$, Directrix: $x = -1$

c. Vertex: $(-1, 2)$, Focus: $(-1, 4)$, Directrix: $y = 0$

d. Vertex: $(2, -1)$, Focus: $(4, -1)$, Directrix: $x = 0$

$4p = 8 \Rightarrow p = 2$. Vertex is $(h, k) = (-1, 2)$. Opens right: Focus is $(h + p, k) = (1, 2)$, directrix is $x = h - p = -1 - 2 = -3$.

28 [Foundational] • Target: Parabolas with Horizontal Axis of Symmetry

Find the equation of a parabola with vertex $(3, -1)$ and focus $(1, -1)$.

a. $(y + 1)^2 = -8(x - 3)$ ✓

b. $(x - 3)^2 = -8(y + 1)$

c. $(y + 1)^2 = 8(x - 3)$

d. $(y - 3)^2 = -4(x + 1)$

Horizontal axis $y = -1$. Opens left because focus $x = 1$ is to the left of vertex $x = 3$. $p = 1 - 3 = -2$. $(y + 1)^2 = 4(-2)(x - 3) = -8(x - 3)$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Parabolas with Horizontal Axis of Symmetry**

Convert $y^2 + 4y - 4x + 8 = 0$ to standard form.

a. $(y+2)^2 = 4(x-1)$ ✓

b. $(y-2)^2 = 4(x+1)$

c. $(y+2)^2 = 4x - 8$

d. $(x-1)^2 = 4(y+2)$

$(y^2 + 4y + 4) = 4x - 8 + 4 = 4x - 4 \Rightarrow (y+2)^2 = 4(x-1)$.

30 **[Intermediate]** • **Target: Reflective Properties and Applications**

A satellite dish has the shape of a paraboloid. If the cross-section is $x^2 = 20y$ (in inches), where should the receiver be placed?

a. **At** $(0, 5)$**, which is** 5 inches **above the vertex** ✓

b. 20 inches above the vertex

c. 10 inches above the vertex

d. 2.5 inches above the vertex

Signals parallel to the axis reflect to the focus. $4p = 20 \Rightarrow p = 5$ inches above the vertex.

31 **[Intermediate]** • **Target: Reflective Properties and Applications**

A suspension bridge cable hangs in the shape of a parabola with towers 100 m apart and 20 m high, touching the road at the center. Find its equation with origin at the vertex.

a. $y = \frac{1}{125}x^2$ ✓

b. $y = \frac{1}{250}x^2$

c. $y = \frac{1}{50}x^2$

d. $y = \frac{1}{20}x^2$

Vertex at $(0, 0)$. Towers at $(\pm 50, 20)$. $20 = a(50)^2 = 2500a \Rightarrow a = \frac{20}{2500} = \frac{1}{125}$.

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32 [Intermediate] • Target: Reflective Properties and Applications

Why do parabolic headlights use a light source placed exactly at the focus?

a. Light rays reflecting off the parabolic surface emerge in a parallel beam ✓

b. It spreads the light in all directions equally

c. It minimizes heat production

d. It prevents the mirror from tarnishing

By the geometric reflective property of parabolas, all rays emanating from the focus reflect parallel to the axis of symmetry.

33 [Intermediate] • Target: Parabolas with Horizontal Axis of Symmetry

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4 ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

34 [Intermediate] • Target: Reflective Properties and Applications

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. 4 ✓

c. 5

d. 16

The equation of a circle centered at the origin with radius r is given by $x^2 + y^2 = r^2$. In this problem, the equation of the circle is given as $x^2 + y^2 = 25$. This implies that the radius squared, r^2, is equal to 25. The circle passes through the point $P(3, y)$. Substituting the coordinates of point P into the equation of the circle, we get: $3^2 + y^2 = 25$. This simplifies to $9 + y^2 = 25$. To find the value of y^2, we subtract 9 from both sides: $y^2 = 25 - 9$, which gives $y^2 = 16$. Taking t.

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35 [Intermediate] • Target: Circles

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$ ✓

The standard equation of a circle centered at the origin $(0, 0)$ with radius r is $x^2 + y^2 = r^2$. Since the circle passes through the point $(3, 4)$, we can substitute these coordinates into the equation to find r^2. $3^2 + 4^2 = r^2$. This gives $9 + 16 = r^2$, so $r^2 = 25$. Therefore, the equation of the circle is $x^2 + y^2 = 25$.

36 [Intermediate] • Target: Parabolas with vertex at the origin

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. **8** ✓

c. 10

d. 4

The equation of the circle centered at the origin with radius 5 is $x^2 + y^2 = 5^2 = 25$. The chord is perpendicular to the x-axis and intersects it at $(3, 0)$. This means the x-coordinate of the endpoints of the chord, A and B, is 3. Substitute $x = 3$ into the circle's equation to find the corresponding y-coordinates: $3^2 + y^2 = 25$ $9 + y^2 = 25$ $y^2 = 25 - 9$ $y^2 = 16$ $y = \pm 4$ So the endpoints of the chord are $A = (3, 4)$ and $B = (3, -4)$. The length of the chord AB is the distance between these t.

37 [Intermediate] • Target: Parabolas with vertex not at the origin

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5° ✓

b. 81.5°

c. 13.5°

d. 17.1°

In the right triangle, opposite $= 3$, adjacent $= 20$, so the angle is $\tan^{-1}\left(\frac{3}{20}\right) \approx 8.5^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 1.01 Functions and Function Notation

For the parabola $y^2 = -20x$, find the vertex, focus, and directrix.

a. **Vertex** $(0, 0)$**, focus** $(-5, 0)$**, directrix** $x = 5$ ✓

b. Vertex $(0, 0)$, focus $(-10, 0)$, directrix $x = 10$

c. Vertex $(0, 0)$, focus $(0, -5)$, directrix $y = 5$

d. Vertex $(-5, 0)$, focus $(0, 0)$, directrix $x = -5$

Comparing $y^2 = 4px$ to $y^2 = -20x$: $4p = -20$, so $p = -5$. Vertex $(0, 0)$, focus $(-5, 0)$, directrix $x = 5$.

39 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 1.02 Domain and Range

For the parabola $y^2 = -16x$, find the vertex, focus, and directrix.

a. Vertex $(-4, 0)$, focus $(0, 0)$, directrix $x = -4$

b. Vertex $(0, 0)$, focus $(-8, 0)$, directrix $x = 8$

c. Vertex $(0, 0)$, focus $(0, -4)$, directrix $y = 4$

d. **Vertex** $(0, 0)$**, focus** $(-4, 0)$**, directrix** $x = 4$ ✓

Comparing $y^2 = 4px$ to $y^2 = -16x$: $4p = -16$, so $p = -4$. Vertex $(0, 0)$, focus $(-4, 0)$, directrix $x = 4$.

40 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 1.03 Average Rate of Change

For the parabola $y^2 = -12x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-6, 0)$, directrix $x = 6$

b. Vertex $(0, 0)$, focus $(0, -3)$, directrix $y = 3$

c. **Vertex** $(0, 0)$**, focus** $(-3, 0)$**, directrix** $x = 3$ ✓

d. Vertex $(-3, 0)$, focus $(0, 0)$, directrix $x = -3$

Comparing $y^2 = 4px$ to $y^2 = -12x$: $4p = -12$, so $p = -3$. Vertex $(0, 0)$, focus $(-3, 0)$, directrix $x = 3$.

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41 [Challenging] • Target: The reflective property of parabolas • Cross-Lesson: 2.01 Right-Triangle Trigonometry

For the parabola $y^2 = -8x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-4, 0)$, directrix $x = 4$

b. Vertex $(0, 0)$, focus $(0, -2)$, directrix $y = 2$

c. **Vertex** $(0, 0)$**, focus** $(-2, 0)$**, directrix** $x = 2$ ✓

d. Vertex $(-2, 0)$, focus $(0, 0)$, directrix $x = -2$

Comparing $y^2 = 4px$ to $y^2 = -8x$: $4p = -8$, so $p = -2$. Vertex $(0, 0)$, focus $(-2, 0)$, directrix $x = 2$.

42 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 2.02 Radian Measure and Arc Length

For the parabola $y^2 = -4x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-2, 0)$, directrix $x = 2$

b. Vertex $(-1, 0)$, focus $(0, 0)$, directrix $x = -1$

c. Vertex $(0, 0)$, focus $(0, -1)$, directrix $y = 1$

d. **Vertex** $(0, 0)$**, focus** $(-1, 0)$**, directrix** $x = 1$ ✓

Comparing $y^2 = 4px$ to $y^2 = -4x$: $4p = -4$, so $p = -1$. Vertex $(0, 0)$, focus $(-1, 0)$, directrix $x = 1$.

43 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

For the parabola $y^2 = 20x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(0, 5)$, directrix $y = -5$

b. Vertex $(5, 0)$, focus $(0, 0)$, directrix $x = 5$

c. Vertex $(0, 0)$, focus $(10, 0)$, directrix $x = -10$

d. **Vertex** $(0, 0)$**, focus** $(5, 0)$**, directrix** $x = -5$ ✓

Comparing $y^2 = 4px$ to $y^2 = 20x$: $4p = 20$, so $p = 5$. Vertex $(0, 0)$, focus $(5, 0)$, directrix $x = -5$.

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44 [Challenging] • Target: Systems involving circles • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

For the parabola $y^2 = 12x$, find the vertex, focus, and directrix.

a. **Vertex** $(0, 0)$**, focus** $(3, 0)$**, directrix** $x = -3$ ✓

b. Vertex $(3, 0)$, focus $(0, 0)$, directrix $x = 3$

c. Vertex $(0, 0)$, focus $(6, 0)$, directrix $x = -6$

d. Vertex $(0, 0)$, focus $(0, 3)$, directrix $y = -3$

Comparing $y^2 = 4px$ to $y^2 = 12x$: $4p = 12$, so $p = 3$. Vertex $(0, 0)$, focus $(3, 0)$, directrix $x = -3$.

45 [Challenging] • Target: Standard Equation and Features of Circles • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For the parabola $y^2 = 16x$, find the vertex, focus, and directrix.

a. **Vertex** $(0, 0)$**, focus** $(4, 0)$**, directrix** $x = -4$ ✓

b. Vertex $(4, 0)$, focus $(0, 0)$, directrix $x = 4$

c. Vertex $(0, 0)$, focus $(8, 0)$, directrix $x = -8$

d. Vertex $(0, 0)$, focus $(0, 4)$, directrix $y = -4$

Comparing $y^2 = 4px$ to $y^2 = 16x$: $4p = 16$, so $p = 4$. Vertex $(0, 0)$, focus $(4, 0)$, directrix $x = -4$.

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